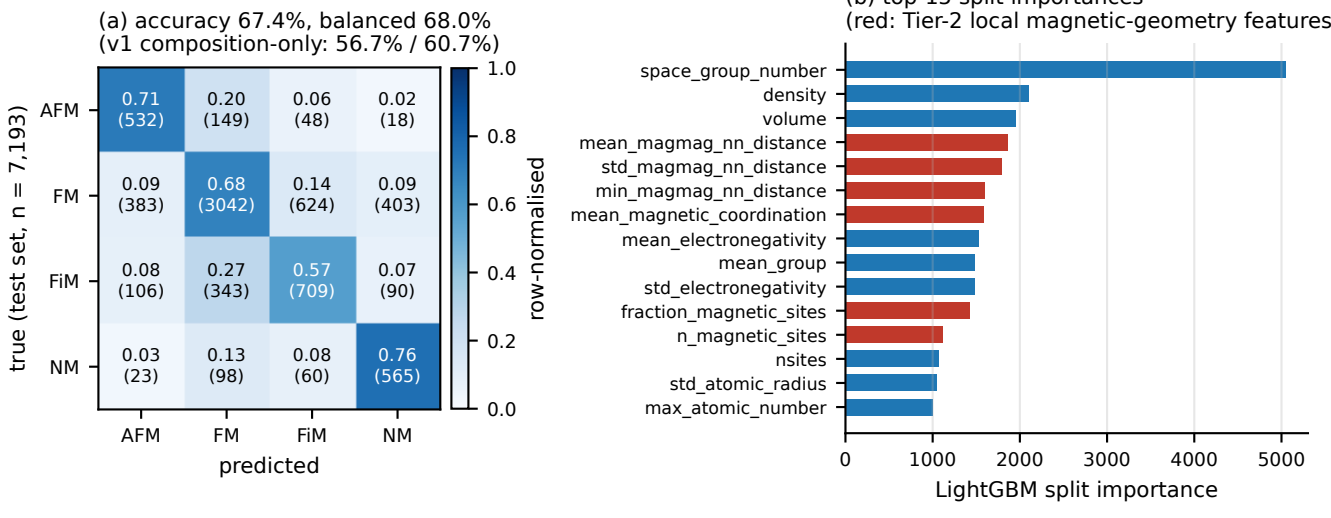
(a) accuracy 67.4%, balanced 68.0%
(v1 composition-only: 56.7% / 60.7%)
true (test set, n = 7,193)
AFM
FM
FiM
NM
0.71 (532)
0.20 (149)
0.06 (48)
0.02 (18)
0.09 (383)
0.68 (3042)
0.14 (624)
0.09 (403)
0.08 (106)
0.27 (343)
0.57 (709)
0.07 (90)
0.03 (23)
0.13 (98)
0.08 (60)
0.76 (565)
AFM
FM
FiM
NM
predicted
row-normalised
1.0
0.8
0.6
0.4
0.2
0.0
(b) top-15 split importances
(red: Tier-2 local magnetic-geometry features)
space_group_number
density
volume
mean_magmag_nn_distance
std_magmag_nn_distance
min_magmag_nn_distance
mean_magnetic_coordination
mean_electronegativity
mean_group
std_electronegativity
fraction_magnetic_sites
n_magnetic_sites
nsites
std_atomic_radius
max_atomic_number
0
1000
2000
3000
4000
5000
LightGBM split importance